HALITE

Confidence-Gated Conflict Resolution over Evolving Maintenance Records

STREAM B · APPLICATION-DERIVED

Shahmeer Atif 23i-0711

Muhammad Umar 23i-0782

Muhammad Usman Al Haq 23i-0585

SUPERVISOR

Syed Muhammad Saad Salman

Maintenance organisations keep records that silently contradict each other — and nothing detects when a new record proves an old one wrong.

A single service report means nothing on its own

10–15 reports a day

Handwritten by technicians, signed by the client

Written for insiders

Terse, abbreviated, jargon-dense field text

Nothing connects them

No link from one report to the next on the same asset

Current workflow: a person re-reads the asset's history, by hand, every time.

SDG 9 · Industry, Innovation and Infrastructure

RECORD 1 11 FEB

Slide light won't extinguish

→ Lubricated girt bar

same aircraft · same fault · same day

RECORD 2 11 FEB

Slide light won't extinguish. CHECK HISTORY.

→ Found debris on proximity sensor

The only link between them is a technician writing an instruction to another human being.

Technician

Inherits a history that is wrong and unmarked

Records supervisor

Reconciles new against old by hand when something looks off

Auditor

Cannot establish what was believed, or when

Everything stores or retrieves. Nothing validates.

WHAT EXISTS	WHAT IT DOES	WHAT IT CANNOT DO
CMMS / MRO platforms	Store work orders, schedule, asset registry	Never checks a new record against stored history
Siemens Industrial KG, IBM GraphRAG	Aggregate everything about an asset for the technician	Retrieval only — nothing is ever marked untrue
RAG assistants over documents	Retrieve top-k by similarity, generate an answer	Ranks by relevant, has no notion of still true
Prior FYP pattern: RAG chatbot over institutional docs	Retrieval wrapper over a document set	Guide §15 weak pattern — capability supplied by the model

A superseded record is maximally similar to its own subject — so retrieval ranks it high, and flags nothing.

We went looking for whoever had already done this

Knowledge graphs for industrial maintenance

Systematic review, 2025 · 270 screened, 76 examined

Three application categories: fault diagnosis, action identification, maintenance organization. Record consistency is not one of them.

KEO — aviation maintenance KG + LLM + RAG

Notre Dame, Oct 2025 · closest work to ours

We counted the full 17-page text: conflict 0 · contradict 0 · supersede 0

PaTeCon — conflict detection on knowledge graphs

2023 · Wikidata, Freebase

It works — on RDF quadruples, needing a hard constraint to violate. Free text has neither.

Boeing + FAA sdr-hazards-classification

Public repository · our exact corpus

Boeing works this dataset as single-record classification. Never record against record.

The complexity is in the constraints

2,160,081 → **45,764** → **462** → **≤ 25**

every record × every record

within-asset — one graph edge, 47× cut

records that reach a model (77.8% never do)

candidates per record

APPLICABLE SEOUL ACCORD CHARACTERISTICS *6 diverse stakeholders — present but limited, and evidenced by POC 0 rather than asserted*

1 Conflicting requirements

Recall competes directly with false-conflict rate. Resolved explicitly at the confidence gate, not hidden in a default.

2 No obvious solution

Which of 2.16 million pairs are worth examining has no standard answer. The naive solution is infeasible, not merely slow.

3 Depth of knowledge

Graph pruning strategy, LLM failure modes, confidence calibration, evaluation design, aviation domain knowledge.

4 Unfamiliar issues

Calibrating abstention against a labelled set, and pre-registering measurement gates, are not routine development work.

5 Beyond standard practice

No framework or template produces conflict adjudication over a record corpus. Neo4j, FastAPI and a hosted LLM solve none of it.

7 Significant consequences

A wrong adjudication silently invalidates a real service record on a regulated aircraft. The system must be able to abstain.

8 Interdependence

Pruning sets classifier load; classification sets adjudication input; confidence sets routing. Each changes the others.

9 Ill-defined requirements

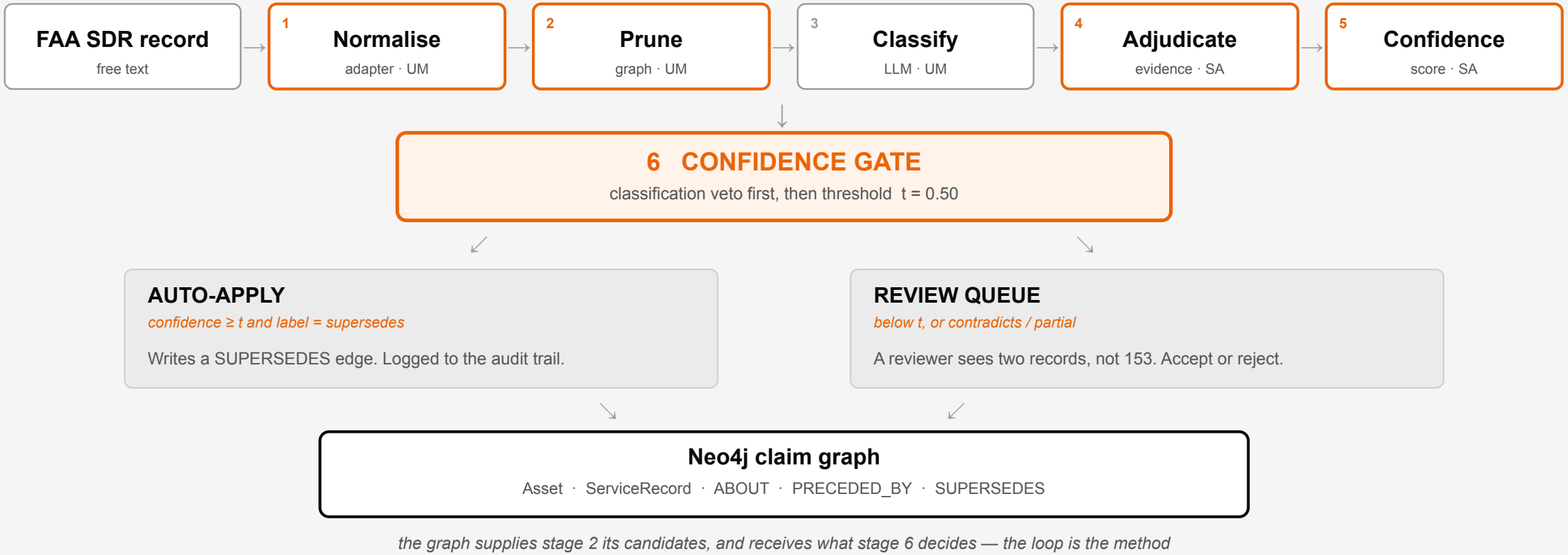
“Does B supersede A?” is not crisply defined. Partial supersession is real; two models labelled one pair differently.

CENTRAL TRADE-OFF recall against false-conflict rate — we hold 0.048 where the recency baseline holds 1.000.

UNCERTAINTY partial supersession is not crisply definable; classifier stability is an open weakness.

WHY ROUTINE DEVELOPMENT IS INSUFFICIENT If the corpus were clean, supersession were annotated by hand, and being wrong cost nothing — this would not be hard.

Proposed methodology



the graph supplies stage 2 its candidates, and receives what stage 6 decides — the loop is the method

CONTRIBUTION

Graph-constrained pruning, conflict classification, **evidence-based adjudication with no user input**, and a **calibrated gate that abstains** — evaluated against author-recorded FAA ground truth.

DEPLOYMENT TARGET

Containerised FastAPI on a VM · React on Vercel · Neo4j managed

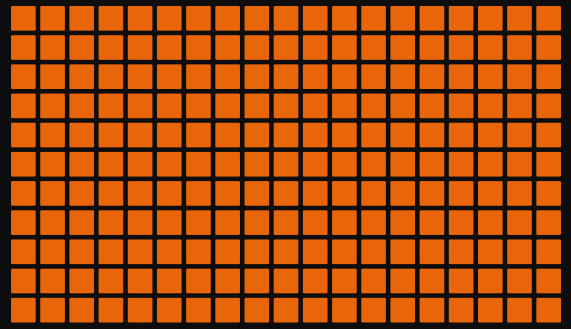
OUT OF SCOPE demand forecasting · airworthiness calls · DERIVED_FROM propagation

Running system, and a baseline run

neo4j · live	2,079	118	141
graph_mode	records ingested	airframes	author-recorded pairs

LIVE — submit a real FAA record with its citation stripped · "it examined 25 of this aircraft's 153 records" · review queue · accept · SUPERSEDES edge written

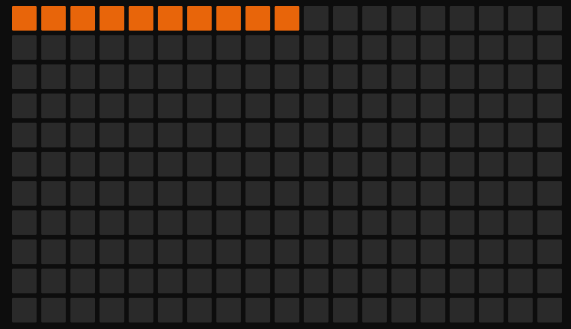
BASELINE — assume most-recent prior



209 / 209 false conflicts on 209 negative pairs

It cannot abstain.

HALITE — t = 0.50



10 / 209 false conflicts on the same pairs

False-conflict rate 0.048 against 1.000.

precision 0.903 · recall 0.660 · F1 0.762 · **false-conflict 0.048 vs 1.000**

N = 350 (141 positive, 209 negative) · openai/gpt-oss-120b

Measuring success, and what can go wrong

ACCEPTANCE CRITERIA

Detection quality	**recall ≥ 0.75 at precision ≥ 0.85**
False-conflict rate	≤ 0.05 on held-out negatives
Threshold	calibrated on a held-out set, not the evaluation set
Conflict latency	p95 ≤ 5s per ingested record at 10k claims
Backend coverage	≥ 80% on core pipeline modules
Auditability	every automatic action reconstructable from the trail

RISKS AND TRIGGERS

Recall too low to be useful	FYP-1 objective → narrow to cited cases
Threshold fitted, not calibrated	held-out calibration set in FYP-1
Classifier instability	self-consistency voting; reported as a limitation
LLM rate limits / cost	cache, retry queue, documented degraded mode
No confirmed external stakeholder	POC 0 formalised; proxy declared now
Neo4j / Groq unavailable	abort to human queue — never silent fallback

TOOL / API / GENAI BOUNDARY

Groq openai/gpt-oss-120b	CORE-ASSIST	Classifies one pair we selected. Sees ≤25 of 11,628. 77.8% of records never reach it.
Neo4j	SUPPORT	Executes our pruning strategy. Replaceable without affecting the contribution.
FastAPI · React · Docker	SUPPORT	Infrastructure and interface.
—	CORE-REPLACEMENT	None. No component supplies pairing, adjudication, calibration or the gate.

AI-assisted SDLC: Level 2. AI usage log maintained per component; every CCP-relevant output verified against labelled data before it is relied on. Each member can explain, justify against a rejected alternative, and modify their own component.

Three owners, one pipeline

Muhammad Umar
23i-0782

Pipeline stages 1–3

- Claim normalisation
- Graph-constrained candidate pairing
- Conflict classification
- Graph schema and indexes

Shahmeer Atif
23i-0711

Pipeline stages 4–6

- Evidence-based adjudication
- Confidence calibration
- Confidence gate and routing
- Review queue and audit trail

Muhammad Usman Al Haq
23i-0585

Adapters, data, evaluation

- Domain adapter
- Corpus acquisition and labelling
- Evaluation harness and baselines
- RBAC, tenancy, deployment

Removal test: remove any one member and you lose a pipeline half, the evaluation basis, or the corpus and deployment. Nobody owns only interface work.

MILESTONES

Proposal

Graph live on 2,079 records; detection end to end; POCs executed

FYP-1 Mid

Recall work; all five classes; RBAC scaffolded; SRS submitted

FYP-1 Final

Threshold calibrated held-out; queue role-restricted; deployed; SDS

FYP-2 Mid

Full harness vs baselines; NFRs measured; ≥80% coverage; load test

FYP-2 Final

Numbers finalised; UAT with practitioners; installable from README

Inclusion, ethics and legal basis

INCLUSION

Knowledge equity

A twenty-year veteran carries an asset's history in their head. A new joiner does not. That gap falls hardest on junior and migrant technicians.

We never ask authors to change

The system consumes records exactly as technicians write them — terse, abbreviated, non-native English. Demanding cleaner writing would penalise the people it should help.

Accessible interface

Light and dark themes; held to WCAG AA contrast.

ETHICS

Automation bias is the real risk

A technician trusting a superseded flag and skipping a check they should have made.

Four guards

We never delete, only mark. Every action shows its rationale, confidence and compared record. Below threshold we abstain. Everything is written to an audit trail.

What Halite is not

A records-integrity tool. It makes no airworthiness determination and issues no maintenance instruction.

LEGAL

FAA SDR is public domain

Works of the US federal government — 17 U.S.C. § 105. No permission required. Source cited.

No personal data

Twelve columns, all aircraft and part attributes. No technician names, no operator personnel, no passengers.

Ground truth is not ours

Labels were authored by FAA technicians. No label engineering was performed — which is why the evaluation means anything.